



Strathmore
UNIVERSITY

SCHOOL OF COMPUTING AND ENGINEERING SCIENCES (SCES)
BACHELORS OF SCIENCE IN NETWORKS AND CYBERSECURITY
ASSIGNMENT 2
DATA AND VOICE COMMUNICATION

DATE:23/06/2025

DUE DATE:30/6/2025

Instructions

1. This examination consists of **TWO** questions.
2. Answer **BOTH** questions.

QUESTION ONE

- a) Explain the need of a modem in a network
(3 marks)
- b) Explain the importance of each of the following line encoding properties
 - i. Synchronization
 - ii. Cost and complexity(4 marks)
- c) For the data sequence, 100001101000011 obtain the line code for each of the following encoding schemes
 - i. Manchester
 - ii. Differential Manchester
 - iii. Bipolar AMI
 - iv. HDB3(8 marks)

QUESTION TWO

- a) For each of the following switching techniques, write down **two distinguishing** features
 - i.Circuit switching
 - ii.Message switching
 - iii.Datagram packet switching(6 marks)
- b) Describe each of the following transmission impairments
 - i. Attenuation distortion
 - ii. Delay distortion
 - iii. Noise(6 marks)